CALIFORNIA STATE UNIVERSITY
LONG BEACH

CECS 341: Computer Architecture & Organization

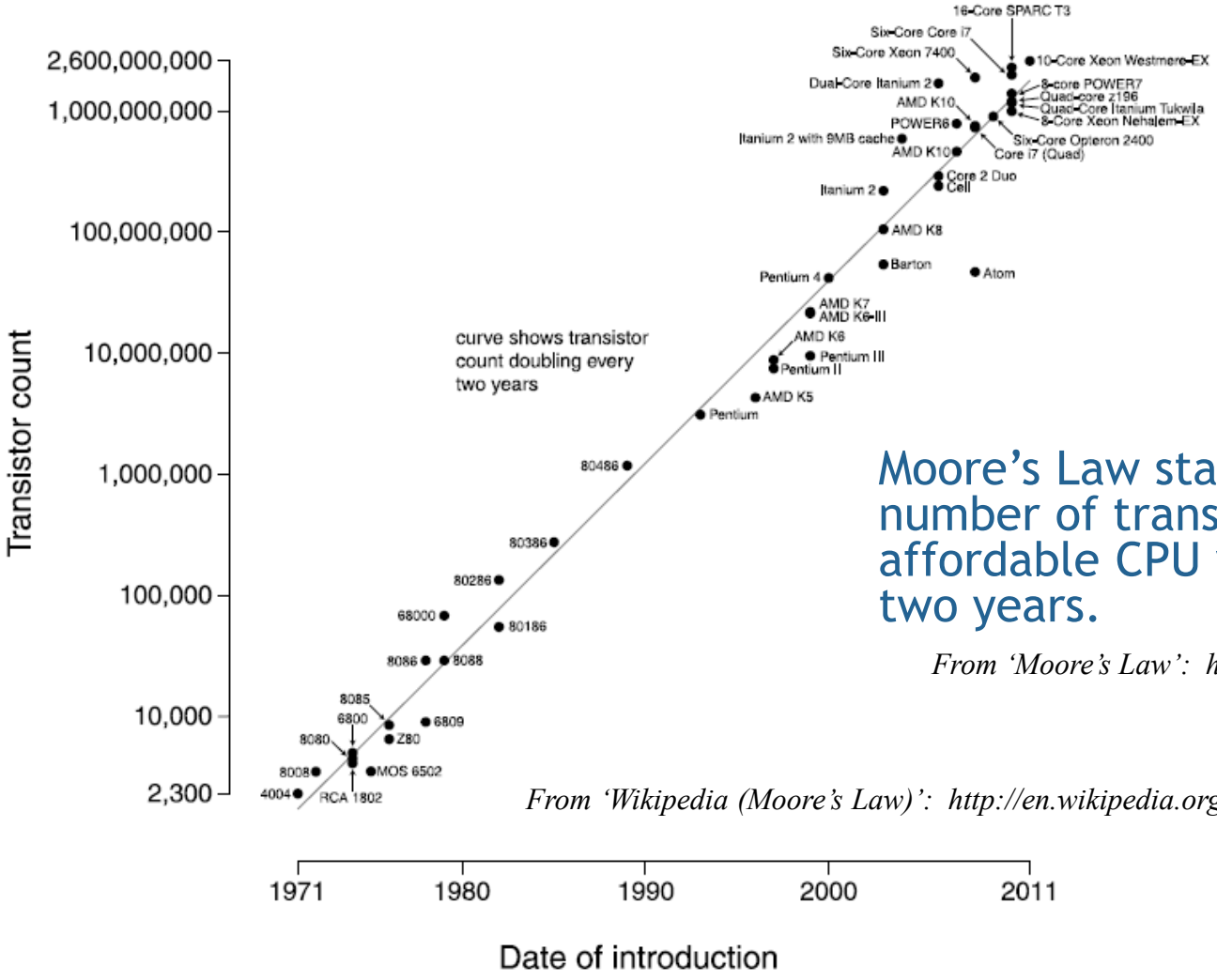
Hossein Sayadi, PhD

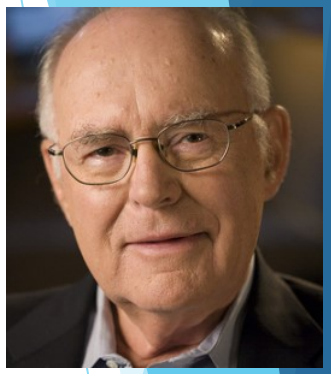
Department of Computer Engineering & Computer Science
California State University, Long Beach

Fall 2025

Introduction to Computer Architecture

Moore's Law





Gordon Moore

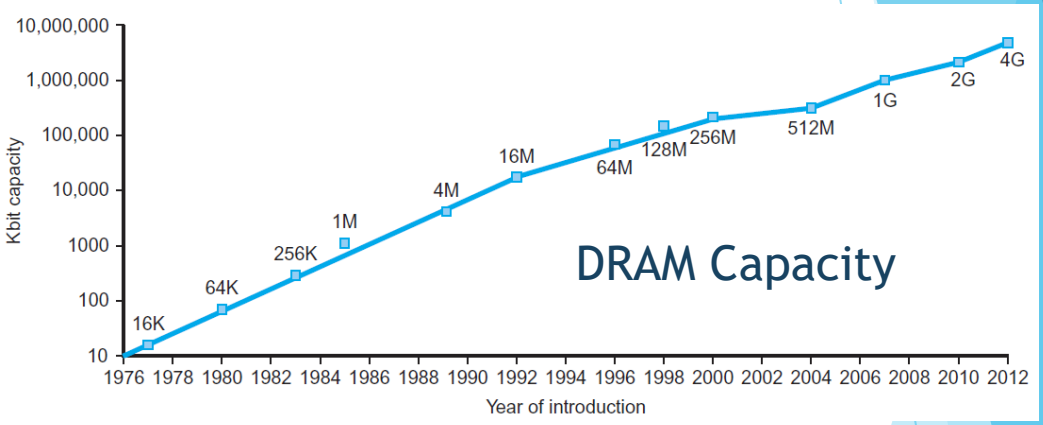
Moore's Law stated that the number of transistors on an affordable CPU would double every two years.

From 'Moore's Law': http://www.mooreslaw.org/

From 'Wikipedia (Moore's Law)': http://en.wikipedia.org/wiki/Moore's_law

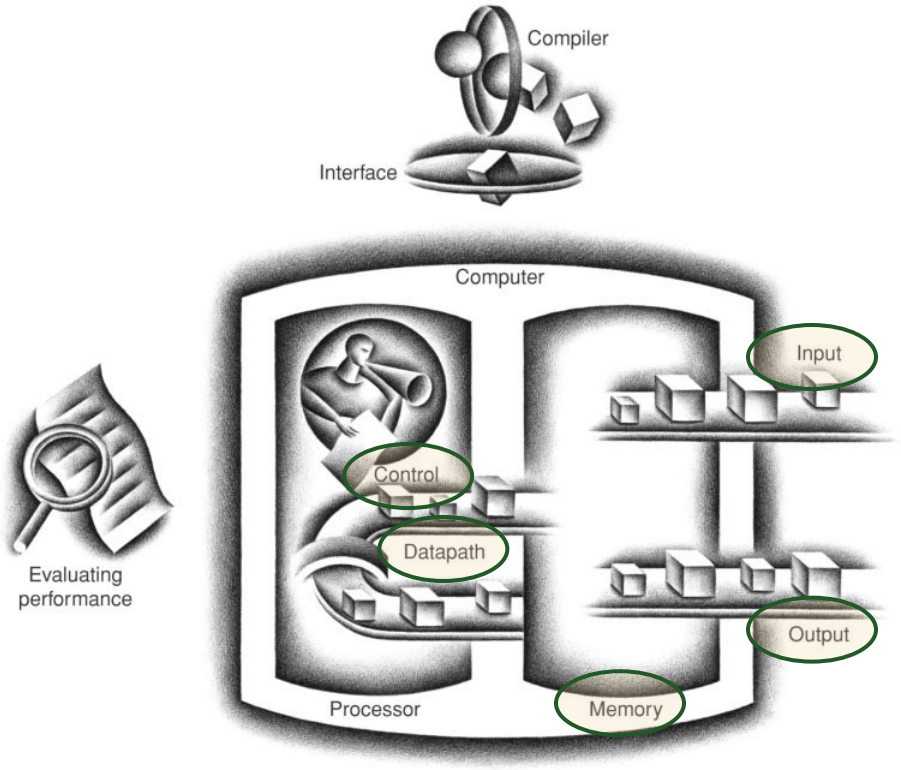
Technology Trends

- Electronics technology continues to evolve.
  - Increased capacity and performance.
  - Reduced cost.



Year	Technology	Relative performance/cost
1951	Vacuum tube	1
1965	Transistor	35
1975	Integrated circuit (IC)	900
1995	Very large scale IC (VLSI)	2,400,000
2013	Ultra large scale IC	250,000,000,000

Classical Components of a Computer

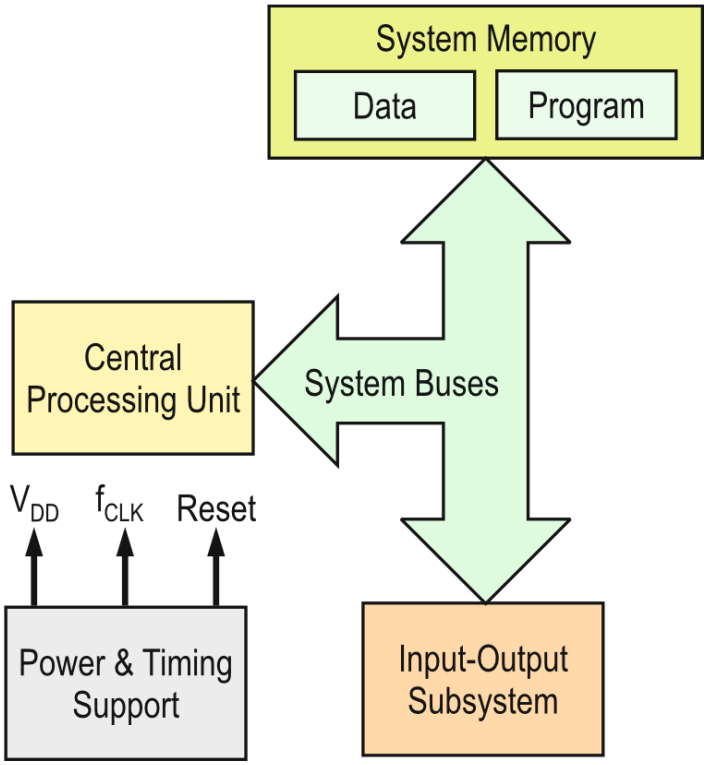


Five classic components

- Same components for all types of computers, including
  - Desktops and laptops
  - Servers
  - Embedded systems
- Input/Output includes
  - User-interface devices
    - Keyboard, mouse, monitor
  - Storage devices
    - Hard disk, CD/DVD, flash
  - Network adapters
    - For communicating with other computers.

Microcomputer System

Architecture of a computer system:

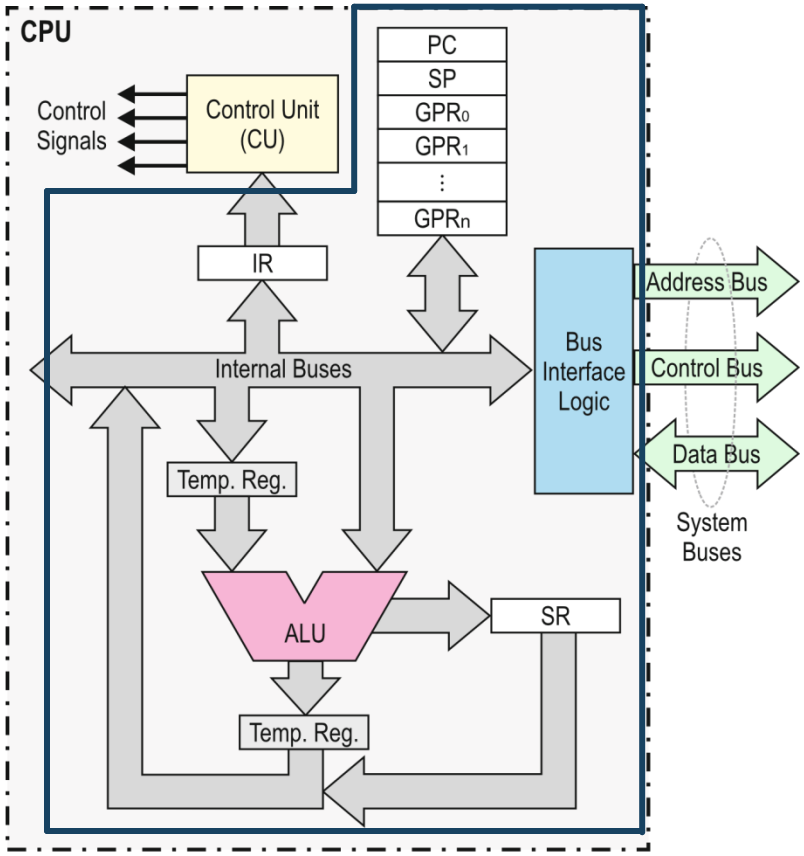


- In general, a computer system is composed of
  - Central Processing Unit (CPU)
    - Datapath
    - Control unit
  - Memory
    - Program
    - Data
  - Input/Output
  - System Buses
    - To connect the CPU to memory and I/O
    - Include the address, data, and control buses

"Introduction to Embedded Systems using Microcontrollers and the MSP430"; Jimenez, Palomera, Couvertier; p. 82

Central Processing Unit (CPU)

Architecture of a simple CPU:

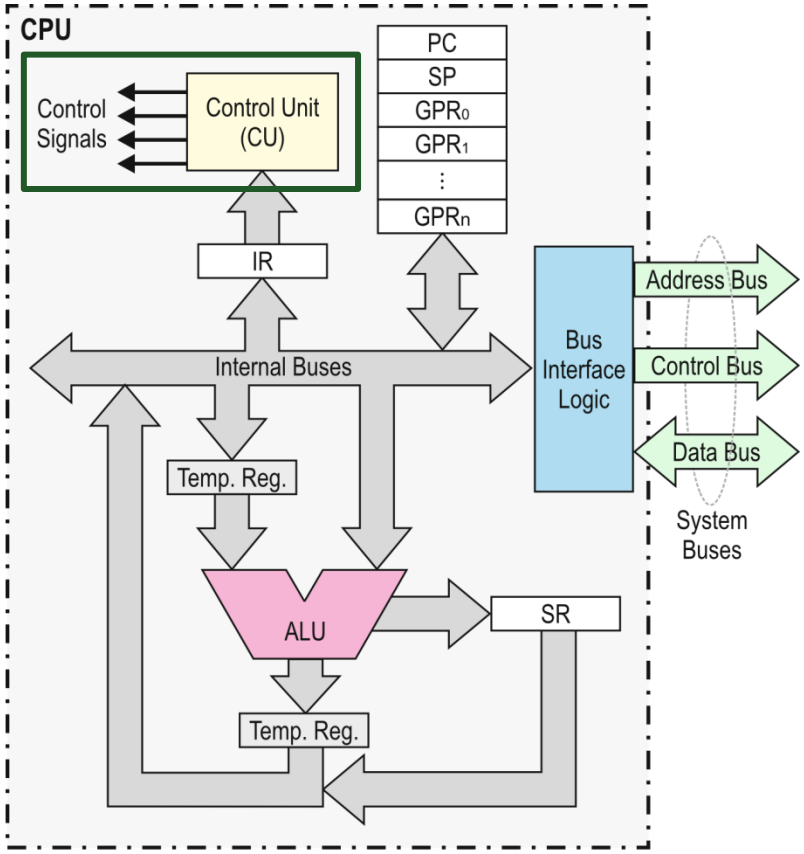


- Datapath
  - Consists of the functional units needed to execute the assembly language instructions in the CPU's instruction set.
  - Functional Units include:
    - Arithmetic and Logic Unit (ALU)
    - Register File
    - Other special-purpose registers
    - Multiplexers
  - Connects to memory and I/O via the Bus interface logic and the System buses.

"Introduction to Embedded Systems using Microcontrollers and the MSP430"; Jimenez, Palomera, Couvertier; p. 87

Central Processing Unit (CPU)

Architecture of a simple CPU:

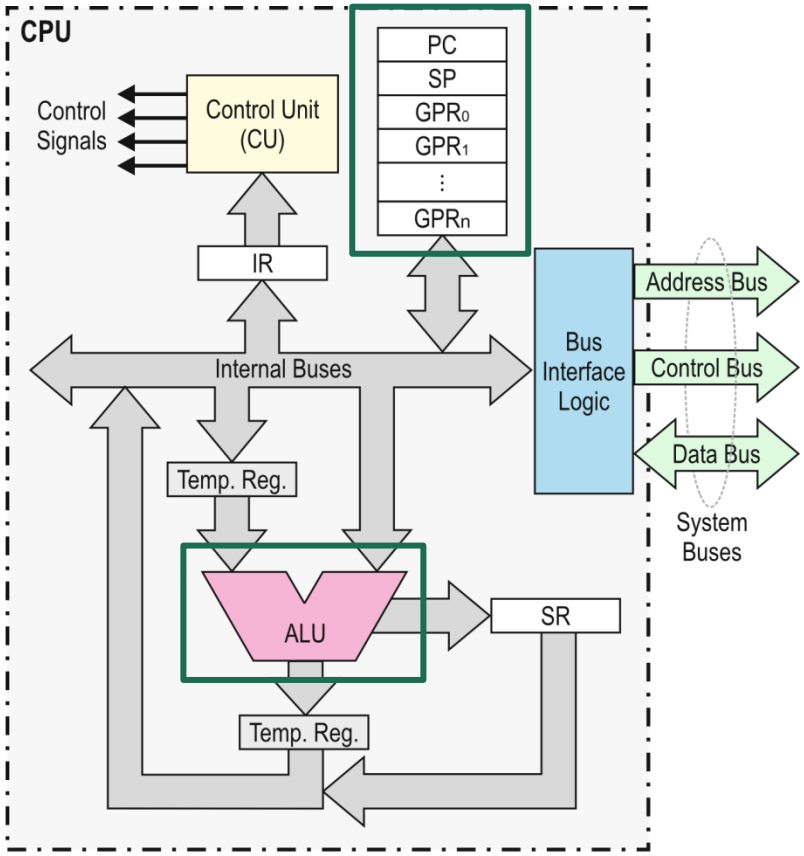


- Control Unit
  - Decodes the assembly language instructions.
  - Generates control signals.
    - Asserts a specific set of control signals based on the AL instruction decoded.
  - Controls the actions of the datapath.
    - The actions needed to execute the decoded AL instruction.
  - Controls communication between the CPU and the memory and I/O.
    - Via the bus interface logic.

"Introduction to Embedded Systems using Microcontrollers and the MSP430"; Jimenez, Palomera, Couvertier; p. 87

Central Processing Unit (CPU)

Architecture of a simple CPU:

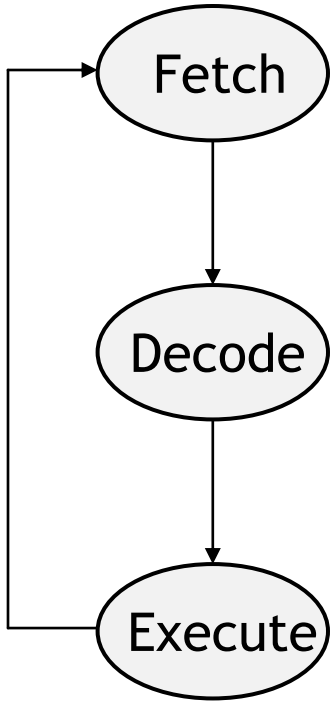


- ALU
  - Performs arithmetic and logic operations.
  - Two source operands.
  - One destination operand.
- Register File
  - General-purpose registers.
  - Special-purpose registers.
  - Programmer accessible.
  - Stores assembly language instruction operands.

"Introduction to Embedded Systems using Microcontrollers and the MSP430"; Jimenez, Palomera, Couvertier; p. 87

Instruction Cycle

- Instructions are executed by the processor.
- Each is processed according to the steps in the *instruction cycle*.
  - Aka. *fetch-execute cycle* and *CPU cycle*.



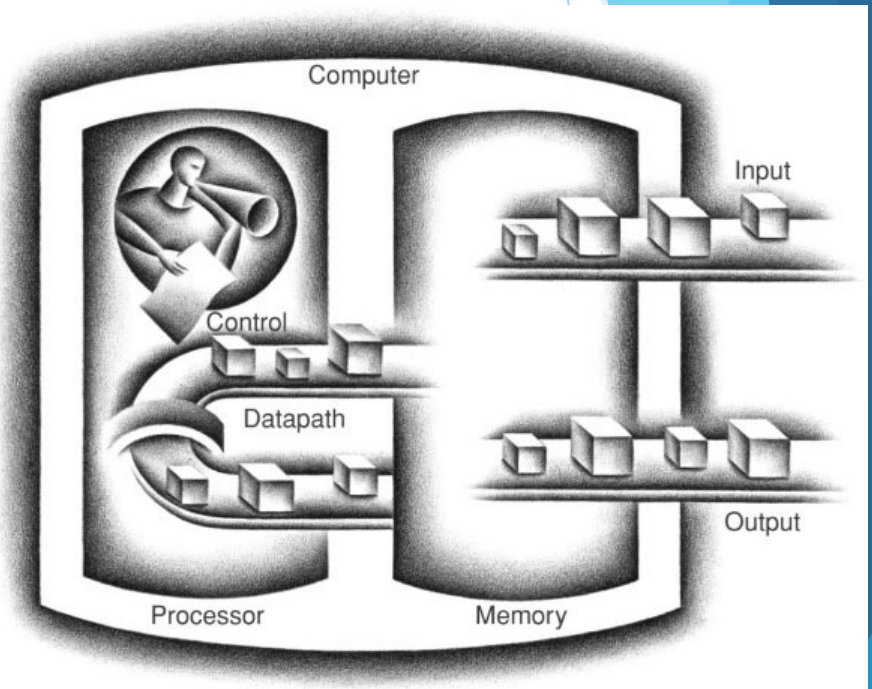
Instruction is fetched (or read) from program memory.

Instruction is decoded by the control unit; Appropriate control signals are generated.

Functional units in the datapath perform the actions needed to implement the instruction.

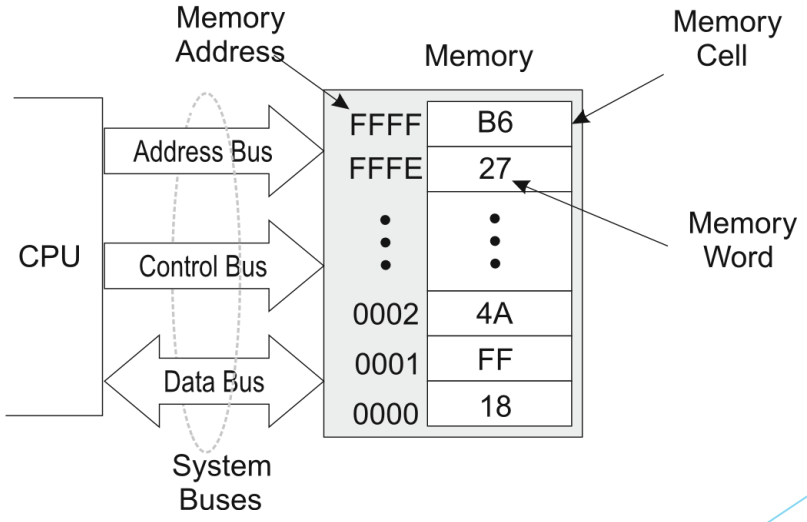
The Memory

- Memory
  - Technologies include:
    - SRAM and DRAM
    - Magnetic media (hard disk)
    - Optical media (CD/DVD)
    - Solid state (flash)
  - Composed of a set of locations
    - Each has a unique address
    - Each can store unique data
  - Organized hierarchically
    - Cache
    - Main memory
    - Secondary memory



Memory Structure

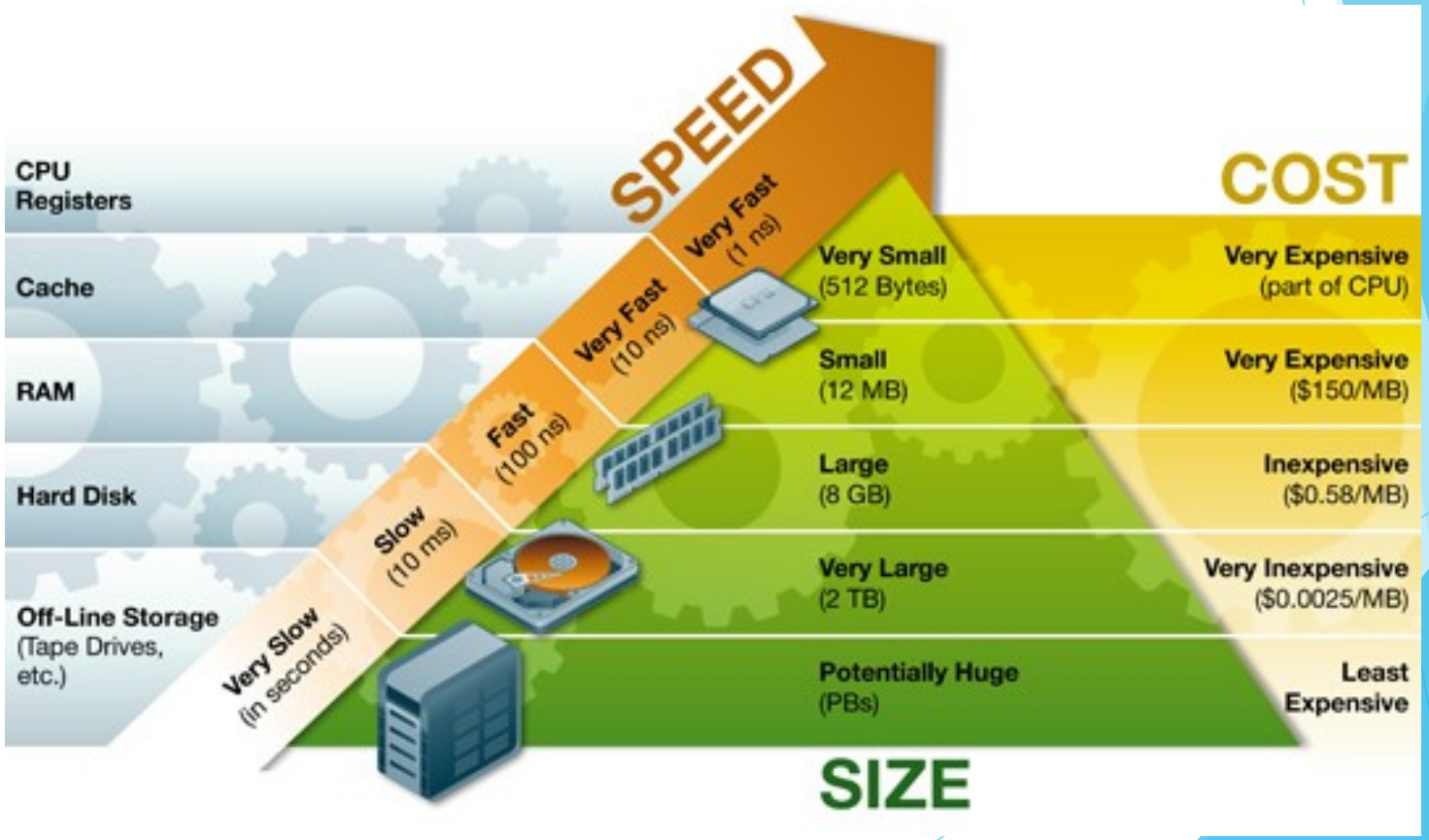
- The memory is organized as an array of cells or locations.
- Each memory location is identified by an address.
- Memory cells may store instructions or data.
  - Each cell contains a memory “word”.



“Introduction to Embedded Systems using Microcontrollers and the MSP430”; Jimenez, Palomera, Couvertier; p. 98

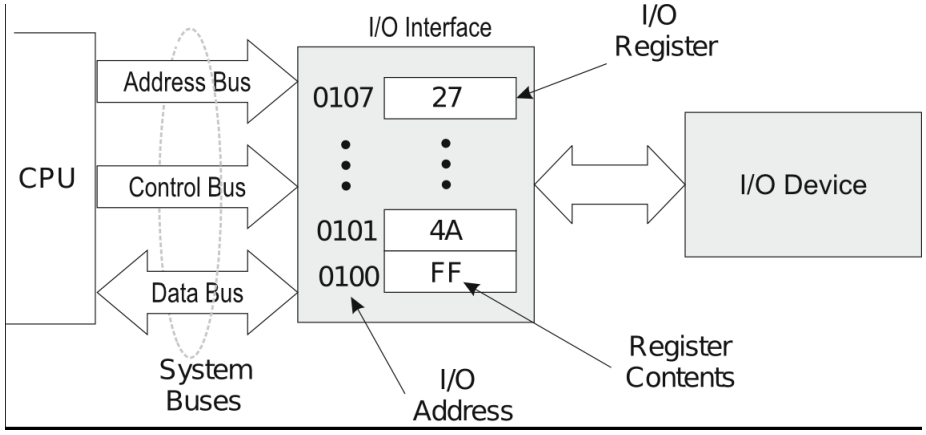
Memory Hierarchy

- The memory system is organized hierarchically:



I/O Device Interface

- Serves as a bridge between an I/O device and the system buses.
- Has one or more registers each with its own address.
  - Data, Control, and Status registers.
- Accessed like memory (read/write).



"Introduction to Embedded Systems using Microcontrollers and the MSP430"; Jimenez, Palomera, Couvertier; p. 111